PROMPT ENGINEERING EXERCISES FOR FINANCE-RISK DOMAIN

Exercise Set 1 — Narrative interpretation of Indian market moves

Prompt 1

NIFTY fell by 2.4% today, while India VIX spiked by 18% and USD/INR weakened sharply.

Explain in 6–8 lines what macro or domestic factors could realistically drive these moves in the Indian market context.

Prompt 2

Write an executive summary (100–140 words) explaining why a sudden spike in long-term G-Sec yields can trigger selling pressure in Indian banking stocks.

Tone: CFO audience in a large private-sector bank.

Prompt 3

Explain the difference between foreign portfolio investor (FPI) risk sentiment vs domestic mutual fund flows in driving NIFTY market rallies and sell-offs.

Give 3 real Indian market examples.

Learning outcome:
LLM already understands **Indian market dynamics** without extra training.

Exercise Set 2 — VaR, P&L, PTA & regulatory alignment in India

Prompt 1 — Policy interpretation

Explain how RBI expects banks to use VaR and Stressed VaR in market risk capital calculations under Basel III in India.

Keep it simple and include references to typical 10-day holding period assumptions.

Prompt 2 — Risk analyst narrative

You are a market risk analyst in an Indian universal bank.

Write a daily commentary for a VaR breach caused by widening credit spreads in PSU issuers after a negative rating outlook revision.

Prompt 3 — PTA interpretation

In the context of FRTB implementation in India, explain why a trading desk may pass backtesting but fail P&L Attribution tests.

Give 2 realistic examples relevant to the Indian market.

Learning outcome:
Participants observe the model can already **align with RBI/Basel expectations**.

Exercise Set 3 — Indian derivatives & Greeks interpretation

Prompt 1

Explain what Gamma risk means to a trader short Bank Nifty index options before weekly expiry. Include a numerical illustration.

Prompt 2

Interpret the following risk statement in simple language:

"The portfolio is long Vega in long-dated USD/INR options."

Who should be concerned and why?

Prompt 3

Why does Theta (time decay) accelerate for deep out-of-the-money short options during expiry week on NSE? Explain in 2 paragraphs.

Learning outcome:
Analysts see the model already understands **Indian derivative products & expiry structure** (e.g., weekly Bank Nifty).

Exercise Set 4 — Stress testing & tail risk with Indian examples

Prompt 1

Design 3 multi-asset stress scenarios specifically for the Indian market involving G-Sec yields, INR, NIFTY, oil prices, and corporate credit spreads. For each scenario provide macro narrative and likely portfolio pain points.

Prompt 2

Explain why Expected Shortfall (ES) is more relevant than VaR for Indian banks with high concentration in long-duration government securities during periods of sharp rate volatility.

Prompt 3

Describe a historically plausible contagion pathway in India:

sharp oil price rise → INR depreciation → imported inflation → tightening cycle → equity correction → corporate credit widening.

Learning outcome:
Participants see **deep contextual understanding of Indian economic linkages**.

Exercise Set 5 — RBI/SEBI regulatory tone transformation

Prompt 1 (regulator tone)

Write a 160-word explanation of a VaR backtesting exception in the tone of an internal memo being submitted to RBI during a supervisory review.

Prompt 2 (CFO tone)

Write a 160-word explanation of a VaR backtesting exception in the tone of the CFO, focusing on business implications and capital efficiency rather than methodology.

Prompt 3 (trader tone)

Write a 160-word explanation of a VaR backtesting exception for the Head of FICC Trading, focusing on risk drivers and potential hedging actions.

Learning outcome:
Demonstrates the model's ability to **context-switch for different stakeholders** — a major pain point in market risk reporting.

Risk oversight & policy reasoning

These show that the model already grasps Indian regulatory frameworks and their objectives.

Prompt 1

What controls does RBI expect banks to implement for Prudent Valuation Adjustments (PVA) on Level 3 assets? Explain in bullet points.

Prompt 2

SEBI has proposed tighter margin and disclosure norms for derivatives trading by retail investors. Explain how this indirectly affects volatility and liquidity in Bank Nifty and NIFTY options markets.

Prompt 3

Explain the governance expectations around model risk management under Basel Pillar 2 for Indian banks. Provide 6–8 actionable bullet points.

Prompt / pre-trained LLM gotchas

Below are **prompt-engineering exercises designed specifically to expose the weaknesses and risks of relying on a base *pre-trained LLM (without RAG or fine tuning)*** especially in **Indian financial markets, risk management and regulatory contexts**.
These are intentionally chosen so that the LLM will *sound confident* but is likely to be **inaccurate, outdated, overly generic, or hallucinatory**.
This creates the *aha!* moment that motivates **exploration of mitigation strategies**

Category 1 — Outdated / Inconsistent / Time-sensitive Knowledge

These prompts often trigger incorrect or outdated information from pre-training.

Prompt A

What are the current RBI regulatory requirements for calculating market risk capital under Basel III for Indian banks? List the exact holding periods and confidence levels required.

Expected issue:
Model may generalize Basel global rules, mention ES incorrectly for India, or invent confidence levels/holding periods that are not in the actual circular.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

Prompt B

Explain the latest SEBI rules on margin requirements for trading Bank Nifty options for hedge funds and proprietary desks.

Expected issue:
Model may hallucinate exact thresholds, mention 2020-era margin reforms, or mix retail/institutional rules.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

Prompt C

What are the latest RBI deadlines for full FRTB adoption by Indian banks?

Expected issue:
Model will almost certainly fabricate timelines.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

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Category 2 — False regulatory permissions (dangerous hallucinations)

These highlight **why out of box pre-trained answers cannot be trusted** in a regulatory environment.

Prompt D

If a bank passes VaR backtesting in the green zone for 12 months, can it temporarily stop calculating Stressed VaR to reduce capital requirements?

Expected issue:
Model may generate *plausible-sounding but unsafe* interpretations of Basel.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

Prompt E

Is it allowed under RBI guidelines to exclude liquidity add-ons if internal liquidity stress models show stable market depth?

Expected issue:
Model is likely to produce a persuasive—but completely incorrect—justification.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

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Category 3 — Mixing global rules with Indian ones

These show the danger of models assuming "Basel = India".

Prompt F

In India, how does the ECB (European Central Bank) guidance on FRTB affect reporting requirements for trading desks?

Expected issue:
Model is likely to fabricate a link between ECB and RBI/SEBI reporting.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

Prompt G

Does SEBI require the same modellable vs non-modellable risk factor (MRF/NMRF) classification thresholds as the US Federal Reserve?

Expected issue:
LLM may align India with Fed rules, which is false.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

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Category 4 — Confident but unverifiable claims

These highlight how pre-training causes **assertiveness without sources**.

Prompt H

How many VaR exceptions does RBI allow in the amber zone before capital multipliers are applied? Give exact numbers.

Expected issue:
Model will likely:

- invent 5–9 numbers from Basel tables,
- confuse internal bank policies with regulation,
- present them *as if factual*.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

Prompt I

What is the exact RBI formula for computing Expected Shortfall under FRTB for market risk?

Expected issue:
There is **no RBI-specific formula**, but the model may invent one.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

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Category 5 — Ambiguous questions that tempt the model to guess

These show why internal policy grounding is needed.

Prompt J

Should the P&L Attribution test in India ignore P&L from new trades booked during the day?

Expected issue:
Model will confidently pick a side and "justify" it.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

Prompt K

If the number of VaR breaches is caused by interest rate volatility, should the trading desk still remain IMA-eligible?

Expected issue:
There is **no one-line rule**, but the model may state one anyway.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations

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Category 6 — MBA-style synthesis questions that *sound right* but require grounding

These show the model inventing arguments unless anchored in documents.

Prompt L

Write 5 bullet points explaining how RBI recently changed Expected Shortfall rules to support capital efficiency for Indian banks with large G-Sec holdings.

Expected issue:
A change like this does not exist — so the model must say **"not in documents"** under RAG.

Validate the answer, ask ChatGPT or copilot and see if it is any better. Document your observations